Risk Management Report

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Part I

1. Individual VaR Computation

We computed the 1-day 99% VaR for each equity position assuming normally distributed returns. Volatilities were estimated using both 125 and 250 past daily returns as of November 21, 2022. The 1-day 99% VaR values based on 125 and 250-day volatilities were CHF 85,383.88 and CHF 89,001.70 for NESN, CHF 192,121.47 and CHF 207,529.49 for SREN, and CHF 152,782.84 and CHF 163,850.54 for NOVN, respectively. The use of 250 days smooths out short-term shocks but may underestimate recent risks, while 125 days reacts more quickly but may be more volatile. While the i.i.d. assumption simplifies VaR computation, it fails to account for volatility clustering, where periods of high or low volatility tend to occur consecutively.

2. Portfolio VaR (Asset-Normal)

Using the Asset-Normal approach, we computed the portfolio VaR accounting for correlations among assets (estimated over 250 returns on Nov 21, 2022). The resulting diversified VaR was CHF 271,339.53, significantly lower than the undiversified VaR of CHF 430,288.19, resulting in a diversification benefit of CHF 189,042.20. This demonstrates how diversification significantly reduces portfolio risk, as accounting for co-movements between assets leads to a much more accurate and typically lower risk estimate than the naive sum of individual VaRs. The short position on Nestle' reduces the value at risk since its positively correlated with the other two stock.

3. Backtesting VaR

Backtesting over the entire period revealed an exceedance rate of 1.96% — nearly double the 1% expected under the 99% confidence level. In 2024 alone, the exceedance rate rose to 2.47%, indicating that the Asset-Normal VaR model significantly underestimated tail risk. This is likely due to increased market volatility or deviations from normality during 2024, which the model fails to capture.

4. RCVaR Analysis

The RCVaR plots over time revealed that NESN, being shorted and relatively stable, contributed negatively and consistently to the portfolio's overall VaR — effectively acting as a hedge. SREN consistently dominated the portfolio's risk contribution, while NOVN's share varied inversely with SREN's.

5. Incremental VaR (iVaR) and Post-Trade VaR

The trade increased exposure to riskier assets (SREN) and reduced the hedging effect of NESN, leading to a higher post-trade VaR (CHF 489,223.28) than the incremental VaR estimate (CHF 476,656.51). This shows that incremental VaR can underestimate risk changes, especially when diversification is affected, since it's a linear approximation.

Part II

6. Systematic market risk

Using historical daily returns and the SMI as the benchmark, we calculated each stock's beta. NESN showed a beta of 0.76, SREN had a higher market sensitivity at 1.10, and NOVN was moderately sensitive at 0.91. These betas quantify each stock's exposure to systematic market risk. Among the three stocks, SREN exhibits the highest systematic risk with a beta of 1.10, indicating higher sensitivity to market fluctuations. Conversely, NESN has the lowest beta at 0.76, reflecting relatively lower exposure to market-wide risk.

7. Volatility models

We estimated the conditional variance of SMI returns using:

- **RiskMetrics (EWMA, $\lambda = 0.94$)**: $\approx 9.25 \times 10^{-5}$
- **GARCH(1,1)**: $\approx 1.43 \times 10^{-4}$

The conditional variance on March 14, 2025, was higher in the GARCH model, indicating greater reactivity.

8. Sharpe Diagonal VaR Comparison

- RiskMetrics-based VaR(99% ; 1): CHF 396,544.51
- RiskMetrics-based VaR(95% ; 1): CHF 279,963.83
- GARCH-based VaR(99% ; 1): CHF 493,190.87
- GARCH-based VaR(95% ; 1): CHF 348,196.99
- Asset-Normal VaR(99% ; 1): CHF 271,339.53

The GARCH-based VaR presents greater reactions to market return shocks compared to the EWMA model. This is because it includes additional variables in the formula, such as the long-term average variance component, the short-term shock response, and the persistence of volatility. Compared to the static asset-normal approach, EWMA and GARCH are dynamic models, meaning that they adjust to market conditions. EWMA creates a smoother forecast and GARCH creates more extreme changes in variance based on market conditions. This can be observed in the plots comparing GARCH and EWMA, as the peaks in early 2022, and late 2024 are higher in the GARCH model.